

Manel Baradad Jurjo

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GitHub

Google Scholar

Machine learning and computer vision researcher specializing in pre- and post-training for generative vision systems. Experience leading multimodal diffusion and image-editing models from data curation and distributed training through evaluation and production deployment. PhD from MIT CSAIL with research spanning generative models, representation learning, and 3D vision.

TECHNICAL EXPERTISE Generative vision · Foundation models · Pre-training · Post-training / fine-tuning · Diffusion models · ControlNet · LoRA · Image editing · Vision-language models · LLMs / RAG · PyTorch · Distributed training · Data curation · Model evaluation

PROFESSIONAL & RESEARCH EXPERIENCE

Amazon Science, BCN, Spain

Sep 2024 – Present

Applied Scientist

- Led the development and **pre- and post-training** of **multimodal diffusion** and image-editing foundation models for product image generation, placement, and marketing creatives deployed on the Amazon platform.
- Developed controllable generation and efficient adaptation methods using **ControlNet models and LoRA adapters**, incorporating **creative-feedback signals** to improve product fidelity, composition, and instruction adherence.
- Built end-to-end **PyTorch** pipelines spanning multimodal data curation, **distributed multi-node, multi-GPU training** on a mid-scale compute cluster, **model evaluation**, and production deployment.
- Amazon AGI: designed and trained neural cartridges for **retrieval-augmented generation (RAG)**, targeting context compression and efficient long-context inference for **large language models**.

Google, GeometricAI, NYC

Jun 2023 – Feb 2024

Research Internship

Researched **wavelet-based volumetric rendering**.

Google, Viscam, Cambridge, USA

Jun 2022 – Sep 2022

Research Internship

Researched **object-centric depth reconstruction** from a single image.

Uber ATG, San Francisco (remote)

Jun 2020 – Sep 2020

Research Summer Internship

Researched **novel-view synthesis** for autonomous-driving simulation.

Earlier Experience

Research Assistant, MIT CSAIL (2017–2018) · Research Assistant, Image Processing Group, UPC (2016–2017) · Software Engineer, Zhilabs (2015–2016) · Research Assistant, Algorithmics and Learning Lab, UPC (2015)

EDUCATION

CSAIL, Massachusetts Institute of Technology (MIT)

2018 – 2024

PhD, Computer Science

Advisor: Antonio Torralba

Polytechnic University of Catalonia (UPC)

2015 – 2017

MS, Electrical Engineering

CFIS, UPC

2011 – 2015

BS, Electrical Engineering · BS, Computer Engineering

SELECTED PUBLICATIONS

PLACID: Identity-Preserving Multi-Object Compositing via Video Diffusion with Synthetic Trajectories

Gemma Canet Tarrés, **Manel Baradad**, Francesc Moreno-Noguer, Yumeng Li. *CVPR 2026*

Separating Knowledge and Perception with Procedural Data

Adrián Rodríguez-Muñoz, **Manel Baradad**, Phillip Isola, Antonio Torralba. *ICML 2025*

A Vision Check-up for Language Models

Pratyusha Sharma, Tamar Rott Shaham, **Manel Baradad**, Stephanie Fu, Adrián Rodríguez-Muñoz, Shivam Duggal, Phillip Isola, Antonio Torralba. *CVPR 2024 (Highlight)*

Procedural Image Programs for Representation Learning

Manel Baradad, Chun-Fu Chen, Jonas Wulff, Tongzhou Wang, Rogerio Feris, Antonio Torralba, Phillip Isola. *NeurIPS 2022*

Learning to See by Looking at Noise

Manel Baradad*, Jonas Wulff*, Tongzhou Wang, Phillip Isola, Antonio Torralba. *NeurIPS 2021 (Spotlight)*

Height and Uprightness Invariance for 3D Prediction From a Single View

Manel Baradad, Antonio Torralba. *CVPR 2020*

Inferring Light Fields From Shadows

Manel Baradad, Vickie Ye, Adam B. Yedidia, Frédo Durand, William T. Freeman, Gregory W. Wornell, Antonio Torralba. *CVPR 2018 (Spotlight)*

Background Prompting for Improved Object Depth

Manel Baradad, Yuanzhen Li, Forrester Cole, Michael Rubinstein, Antonio Torralba, William T. Freeman, Varun Jampani. *arXiv 2023*

HONORS AND AWARDS

"laCaixa" Fellowship (2018; two years of graduate tuition and stipend) · **Fundació Catalunya-La Pedrera Fellowship** (2015; MS tuition) · **CFIS Fellowship** (2012; BS tuition)

Languages English: Fluent · Spanish, Catalan: Native · French: Beginner